NAME

mbwcdview − Display water column data, one swath at a time.

VERSION

Version 5.0

SYNOPSIS

mbwcdview
[
−−**input**=*file* {−**I***file*}
−−**format**=*format_id* {−**F***format_id*}
−−**ping**=*index* {−**P***index*}
−−**swath**=*index* {−**S***index*}
−−**bottom**
−−**wc-bottom**
−−**straight**
−−**export**=*image.png*
−−**size**=*width*x*height*
−−**dump**
−−**verbose** {−**V**}
−−**help** {−**H**}
]

DESCRIPTION

mbwcdview is a Qt6 program that displays the water column data of swath sonar files, read through the MBIO library, as an image of the beamformed amplitudes against across track distance and depth, one swath at a time. Water column data are available for the Kongsberg kmall format (format 261), in the separate *.kmwcd files (water column only), and for the third generation Kongsberg multibeams (EM 122, 302, 710, 712, 2040 ...; formats 58 and 59), in the separate *.wcd files or in the *.all files. The original files are read; no preprocessing is needed. The formats 56 and 57 (for instance EM 3002) are not supported.

When a Kongsberg *.kmwcd, *.kmall, *.wcd or *.all file is opened, **mbwcdview** looks for the other file of the same survey line in the same directory: the same line number (the first four characters of the file name) and the nearest time in the file name. Kongsberg SIS names each file after the time of its first datagram, so the two names may differ by seconds. The water column pings are matched with the survey pings of the *.kmall or *.all by ping number and time. The *.kmall or *.all supplies the navigation, the vessel heading, the transducer depth and the soundings; the *.kmwcd or *.wcd supplies the water column. A *.all without a *.wcd supplies the water column too, if it holds any. Any of these may be missing: without a *.kmall or *.all the navigation is taken from the position records of the *.kmwcd or *.wcd, if present, and depths are given below the transducer. Where water column datagrams were lost in a ping (*.wcd, *.all) the part of the swath they held stays blank.

The water column samples are ray traced through the sound velocity profile logged with the data (**mb_rt**(3), as **mbprocess** traces soundings), from the transducer depth and with the beam angles and the sound speed at the transducer logged with the water column. The soundings then lie on the seafloor echo. With the **Ray trace (SVP)** check box off (or −−**straight**) the samples are placed along straight rays with the sound speed at the transducer; in deep water with a fast surface layer this puts the seafloor echo noticeably deeper than the soundings and the outer beams tens of metres off. No motion during the ping (heave, pitch, roll) is applied.

The amplitudes are shown in dB. For Kongsberg data these are the logged values (0.5 dB steps) minus the gain offset (TVGoffset_dB), the display value defined by Kongsberg. In kmall data the lowest logged value (−128, i.e. −94 dB with a 30 dB offset) is a fill value when the next values are practically absent, and is then drawn as

no data. In *.wcd and *.all data the amplitudes are the logged values minus the TVG offset C, and the lowest value is a valid level, weak samples clipped to it.

The window has controls for the depth range, the amplitude range of the colour scale, a fixed across track scale, the display of the soundings of the matching survey ping (**Bottom**: red amplitude, dark blue phase detections, filled good, hollow flagged) and of the bottom detections recorded in the water column itself (**WC bottom**, orange line), a slider (also mouse wheel and the left and right arrow keys, one swath per step), a replay at a given rate, saving the image at twice the screen resolution with the file name and ping information, and picking positions in the image (**Pick**) into a table that can be saved as text. The image is drawn with OpenGL, so stepping through the pings is fast.

Without a window, **−−dump** prints a summary of a survey line and **−−export** writes an image of one swath.

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OPTIONS

−−input=*file* {**−I***file*}
: The swath file to open. For Kongsberg data either the *.kmwcd or the *.kmall, or the *.wcd or the *.all, of a survey line; the other file is found as described above. The file name may also be given as the last argument.

−−format=*format_id* {**−F***format_id*}
: The MBIO format of the input file. By default it is taken from the file name; *.kmwcd files are read as format 261 and *.wcd files as format 58.

−−ping=*index* {**−P***index*}
: The water column ping (counted from 0) shown first, exported or summarized. Default: 0.

−−swath=*index* {**−S***index*}
: With **−−export**, the swath of the ping (0 is the aftmost swath in multiple swath modes). Default: the first swath.

−−bottom
: With **−−export**, draw the soundings of the matching survey ping.

−−wc-bottom
: With **−−export**, draw the bottom detections recorded in the water column.

−−straight
: With **−−export**, use straight rays instead of ray tracing.

−−export=*image.png*
: Write an image of one swath (PNG or JPEG by the file name) and exit without opening a window.

−−size=*widthxheight*
: The size in pixels of the exported image. Default: 1400x900.

−−dump
: Print a summary of the survey line and exit: the files found, the numbers of pings, matched pings and position records, the amplitude range and fill value, the position, speed, course and heading at the

ping given by −−**ping**, and the depth ratio and across track difference between the bottom detections of the water column and the soundings, for straight and for ray traced geometry.

−−verbose {−V}
Print the program name and version, and with −−**export** what was written.

−−help {−H}
Print the usage message and exit.

EXAMPLES

To view a Kongsberg survey line, open either of its files:
mbwcdview 0080_20250618_223434.kmwcd

To write an image of ping 100 with the soundings and the water column bottom detections:
mbwcdview −I 0030_20250208_100905.kmall −−ping=100 −−bottom −−wc-bottom −−export=ping100.png

To check the pairing of the files and the ray tracing of a line:
mbwcdview −−dump −I 0030_20250208_100905.kmall

SEE ALSO

mbsystem(1), **mbinfo**(1), **mbprocess**(1), **mbio**(3)

BUGS

The soundings of a ping are assigned to the swaths by splitting them into equal blocks, which is right for dual swath data with one receiver; when the receive fans have different numbers of soundings all soundings of the ping are drawn on each swath. Water column data stored inside *.kmall files are not read. Dual head and dual receiver systems, and water column data of EM 2040 scanning mode, have not been tested.